Lecture 13: Confidence Intervals

Problem Set for Lecture 13

These questions will assist in bolstering your understanding of the material in Lecture 13. Emphasis should be placed on having the correct code/output as well as communication. The deliverable should be a knitted RMarkdown document to pdf without any code running off the page. You will be able to present these in office hours as an oral assessment interview along with the problem set for Lecture 11 and Lecture 12.

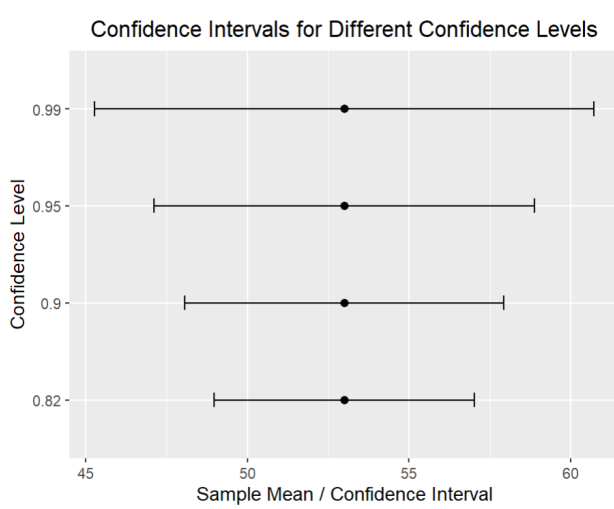
***** First, make sure you have read the Lecture Material's writeup *****

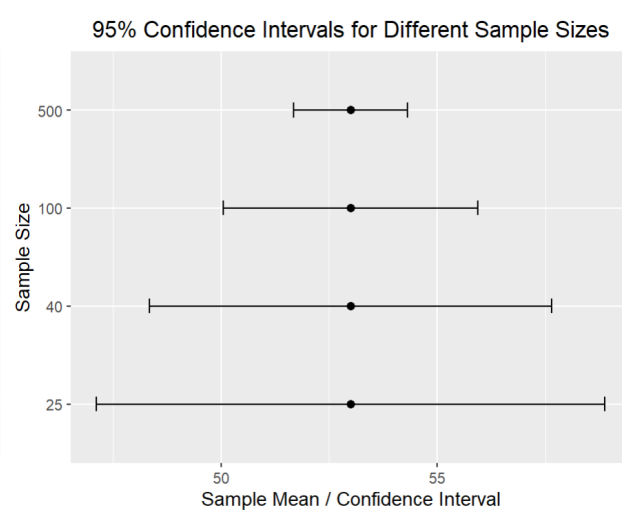
1. A university is studying the average number of minutes students spend on homework each night. Suppose the population standard deviation is known to be 15 minutes, and one sample gave a sample mean of 53 minutes.

   a) Construct an 82%, 90%, 95%, and 99% confidence interval for the true mean number of minutes students spend on homework each night when the sample size is $n = 25$. Create a table and visualization showing the upper and lower bound for the intervals. Additionally, recreate the plot (shown below) that visualizes how the confidence interval changes as the level of confidence is adjusted.

   b) Now keep the confidence level fixed at 95% and construct confidence intervals when the sample size is $n = 25$, $n = 40$, $n = 100$, and $n = 500$. Create a table and visualization showing the upper and lower bound for the intervals. Additionally, recreate the plot (shown below) that visualizes how the confidence interval changes as the level of confidence is adjusted.

   c) In 2–3 sentences, explain how changing the confidence level and sample size affects the width of a confidence interval. Then write one sentence interpreting the 95% confidence interval when $n = 100$ in context.



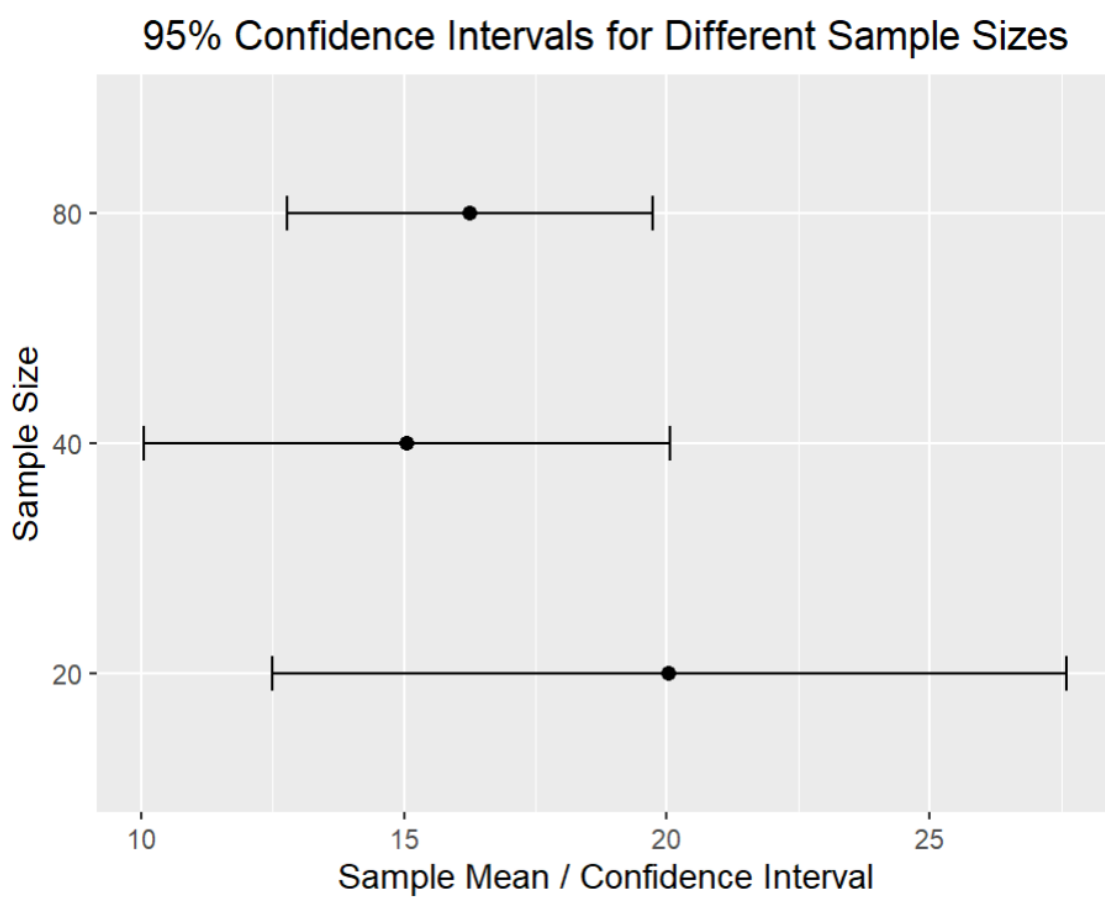


2. Using the `absenteeism` dataset in the `openintro` library, study confidence intervals for a population mean when the population standard deviation is unknown. Use the `days` variable for these questions.

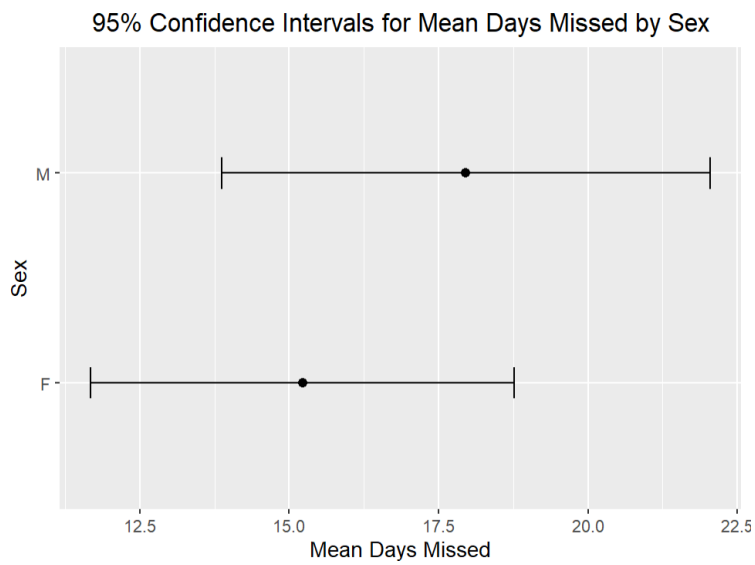
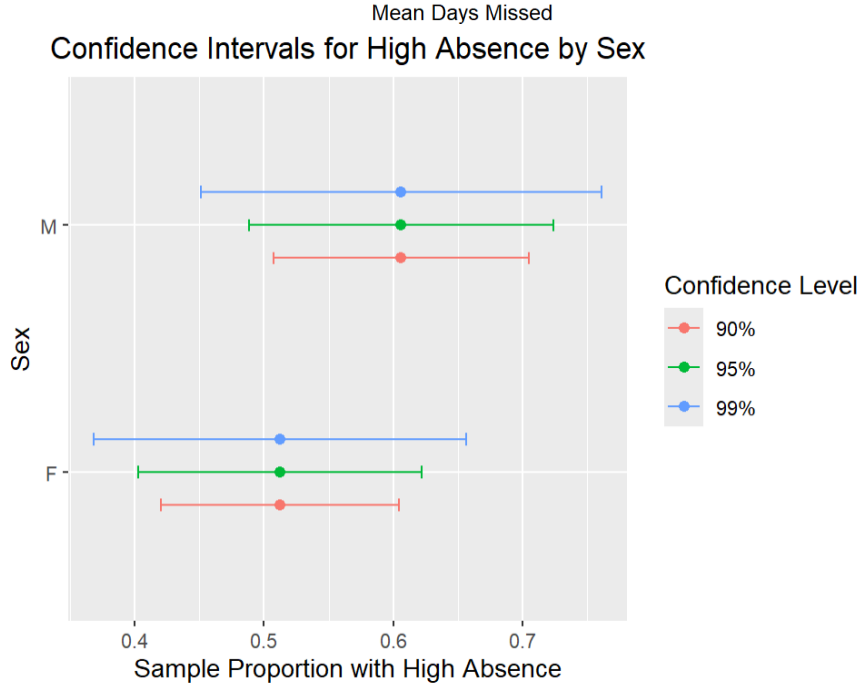
   a) Create a histogram of `days`. Does the distribution appear roughly symmetric, somewhat skewed, or strongly skewed? Briefly describe what you notice.

   b) Using the full dataset, construct a 90%, 95%, and 99% confidence interval for the population mean number of days missed using manual calculations. When presenting your results, have them be in a table that includes the lower and upper bounds for each confidence level. Use the `t.test()` function to verify your 95% confidence interval (you should get exactly the same values!!!).

   c) Set a seed so your work is reproducible. Then take random samples of size $n = 20$, $n = 40$, and $n = 80$ from `days`. For each sample, construct a 95% confidence interval for the population mean number of days missed. Create a table that includes the sample size, sample mean, sample standard deviation, lower bound, upper bound, and width of each 95% confidence interval. Additionally, recreate the plot (shown below) that visualizes how the confidence interval changes as the level of confidence is adjusted. Note that your results will look slightly different then mine!!!



3. Using the same `absenteeism` dataset in the `openintro` library, compute confidence intervals within groups.

   a) Construct a 95% confidence interval for the mean number of days missed for female students and for male students. Create a visualization (seen below) that displays the grouped mean number of days missed along with their 95% confidence intervals using points and error bars.

   b) Define a student as having `high_absence` if they missed at least 10 days of school. Construct a 90%, 95%, and 99% confidence interval for the overall proportion of students with high absence within each sex. Create a visualization (seen below) that displays the grouped sample proportions along with their 95% confidence intervals using points and error bars.